

ComicCraft – AI Comic Story Creator

using Gemini Models

Sub topic **Deployment**

Activity 5.1: Preparing the Application for Local Deployment

- Technology: Python
- Framework: FastAPI
- Template Engine: Jinja2
- AI Model: Gemini
- Deployment Type: Local Deployment
- Objective: Prepare and run the ComicCraft application on a local computer for testing and demonstration.

Slide 2 — Local Deployment Preparation

Preparing the Development Environment

Before running ComicCraft locally, the required software and dependencies are installed.

Requirements:

- Python 3.x
- FastAPI
- Uvicorn
- Jinja2
- Google Gemini API package
- Required project files
- Gemini API Key

Installation Commands

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Slide 3 — Configuring the Application

FastAPI Application Setup

The main application is configured to load routes, templates, static files, and Gemini services.

```
from fastapi import FastAPI
from fastapi.staticfiles import StaticFiles

app = FastAPI(
    title="ComicCraft",
    description="AI Comic Story Creator"
)

app.mount(
    "/static",
    StaticFiles(directory="static"),
    name="static"
)

@app.get("/")
async def home():
    return {
        "message": "Welcome to ComicCraft!"
    }
```

Project Structure

Slide 4 — Running ComicCraft Locally

Starting the FastAPI Server

Use **Uvicorn** to run the application locally.

Server Process

The `--reload` option automatically restarts the server whenever Python source files are modified during development.

Testing

After starting the server, open the local application in a web browser:

FastAPI also provides interactive API documentation at:

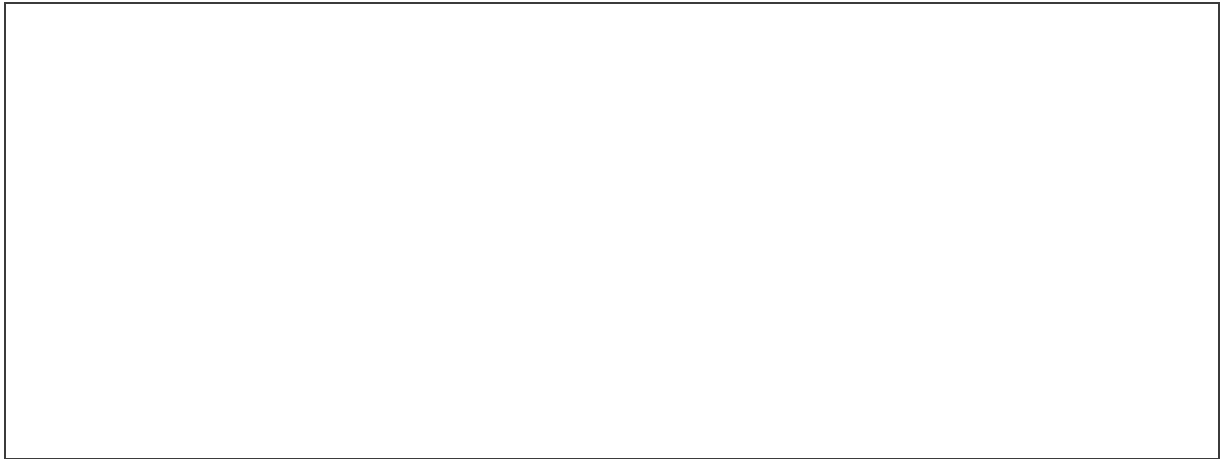
Slide 5 — Milestone 5 Outcome

Activity 5.1: Local Deployment Preparation Completed

Key Achievements:

- Created the Python virtual environment.
- Installed required project dependencies.
- Configured the FastAPI application.
- Connected templates and static files.
- Configured the Gemini API integration.
- Created the required project structure.
- Started the application using Uvicorn.
- Tested ComicCraft through a local web browser.

Final Deployment Workflow



Outcome:

The **ComicCraft AI Comic Story Creator** was successfully prepared for **local deployment and testing using Python, FastAPI, Jinja2, and Uvicorn.**